Task Aware Dynamic Semantic Compression Framework for 6G Deep JSCC

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Motivation / Introduction

6G is shifting from bit-accurate delivery to **meaning- and task-oriented** delivery, yet raw camera and sensor data is too large for the radio link. Separate source and channel coding also suffers a **cliff effect**, failing abruptly below a threshold SNR, while deep JSCC degrades gradually. Receivers differ too: a human operator needs a viewable image, whereas a traffic AI needs only task-relevant semantics. Existing JSCC work usually optimizes either reconstruction or task accuracy, and often ignores payload size.

Scope of the Project

We design a **dual-mode semantic transmitter** for traffic imagery (highway vs urban): one transmitter, two kinds of receiver. **Human mode** sends a 4×64×64 latent (16,384 values) from a frozen pretrained VAE and reconstructs the image. **Machine mode** sends a 128-value semantic vector from a lightweight CNN and classifies it directly. Both travel over a power-normalized AWGN channel (SNR −5 to 25 dB). We measure MSE, PSNR, accuracy and bandwidth savings, and compare JSCC behavior against a conceptual digital outage (cliff) model.

Methodology

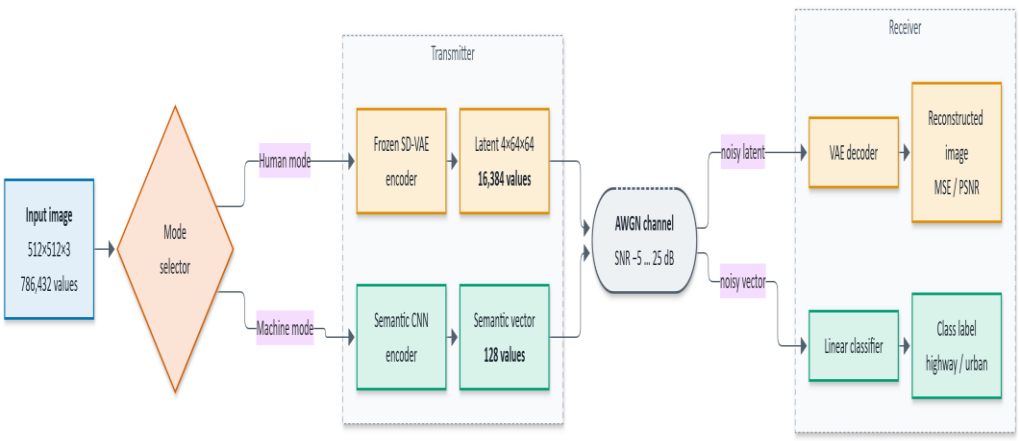


Fig. 1 Overall dual-mode semantic communication system.

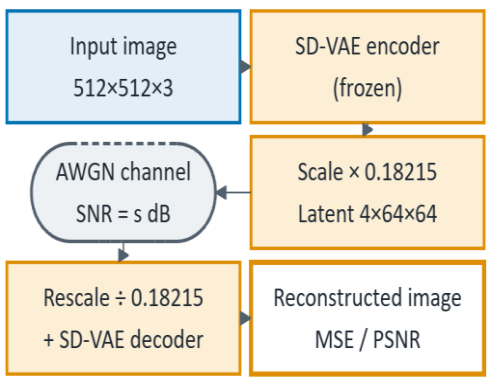


Fig. 2 Human-mode branch.

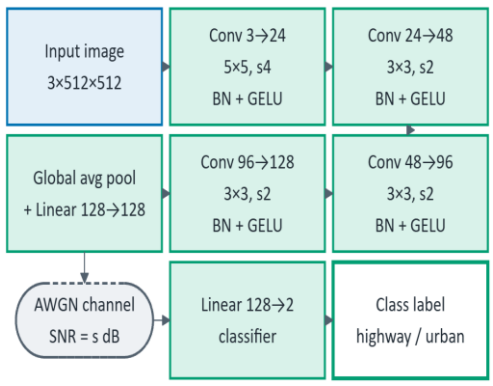


Fig. 3 Machine-mode branch.

Dataset and setup: traffic_data, 200 images, 2 classes (highway, urban), resized to 512×512. Split: 140 train, 60 machine-test, 20 human-test. Python 3, PyTorch, Hugging Face diffusers (AutoencoderKL, sd-vae-ft-mse), AdamW (lr 1e-3, weight decay 1e-4), 80 epochs, cross-entropy, seed 6.

a) Human mode: A frozen VAE encoder maps the image to a 4×64×64 latent (scaled by 0.18215). The latent is corrupted by AWGN and the VAE decoder reconstructs the image. Fidelity is reported as MSE and PSNR.

b) Machine mode: A lightweight CNN (4 Conv + BatchNorm + GELU blocks, global average pooling, Linear 128) produces a 128-value semantic vector. After AWGN, a Linear(128→2) head outputs highway or urban.

c) Channel-robust training: one SNR is drawn at random per batch from {25, 20, 15, 10, 5, 0, −5} dB, so the encoder and classifier learn with the noisy channel in the loop.

$$\sigma^2 = P_{signal} / 10^{(SNR/10)} \qquad PSNR = -10 \cdot \log_{10}(MSE)$$

$$Saving = 1 - N_{tx} / 786{,}432$$

σ^2: AWGN noise power; P_{signal}: average transmitted signal power; N_{tx}: number of transmitted values.

Results

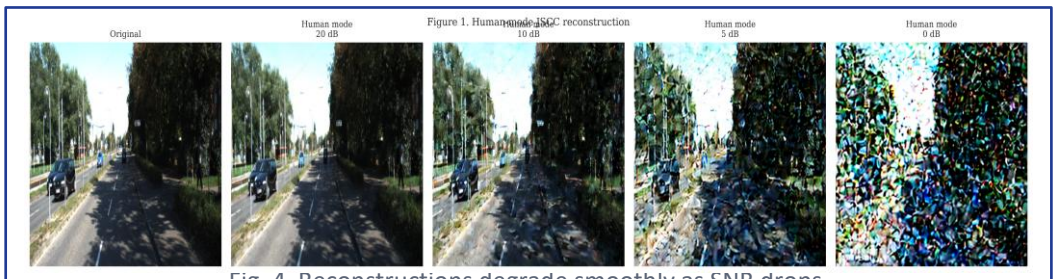


Fig. 4 Reconstructions degrade smoothly as SNR drops.

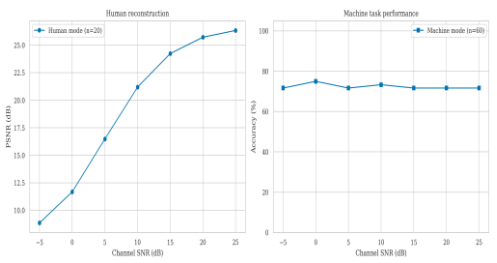


Fig. 5 Human-mode PSNR falls smoothly; machine accuracy stays at 71.7–75.0%.

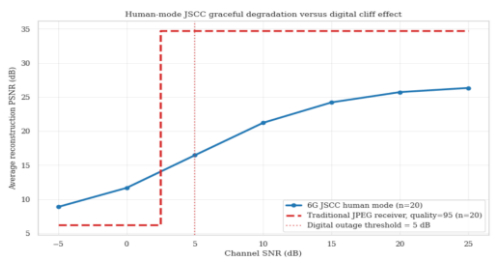


Fig. 6 Graceful JSCC degradation vs a conceptual digital cliff (5 dB threshold).

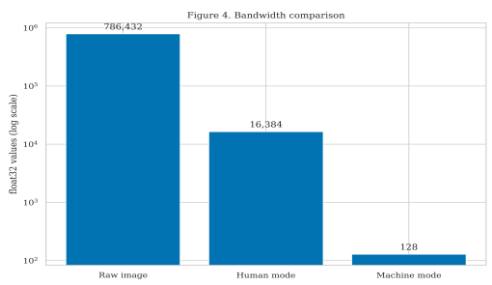


Fig. 7 Payload: raw 786,432, human 16,384, machine 128 values.

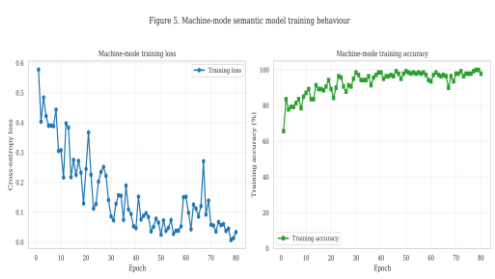


Fig. 8 Machine-branch training over 80 epochs (train accuracy 98%).

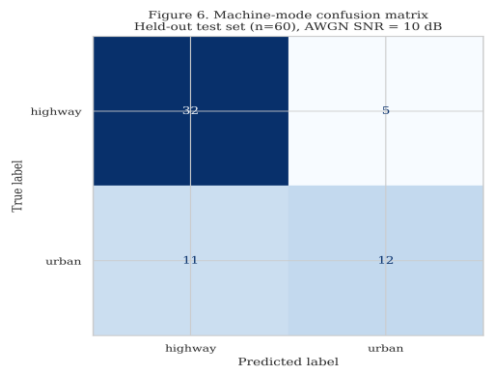


Fig. 9 Held-out set at 10 dB: 73.33% accuracy.

SNR (dB)	PSNR (dB)	Acc. (%)
25	26.34	71.7
20	25.72	71.7
15	24.24	71.7
10	21.19	73.3
5	16.48	71.7
0	11.68	75.0
−5	8.86	71.7

Table 1 Performance vs SNR.

SDGs Addressed:

- **3** — **Good Health and Well-being**: Faster road-incident detection
- **9** — **Industry, Innovation, Infrastructure**: Efficient AI-native 6G links
- **11** — **Sustainable Cities and Communities**: Bandwidth-light traffic monitoring
- **13** — **Climate Action**: Fewer values sent, less energy

Conclusion / Summary

A dual-mode semantic architecture serves human and machine receivers over a noisy 6G-style channel. Payload is cut by **97.92%** (human, 48×) and **99.98%** (machine, 6,144×). Human PSNR degrades gracefully from 26.34 dB to 8.86 dB, and machine accuracy stays stable at 71.7–75.0% across −5 to 25 dB SNR. Future work: larger multi-class datasets, fading and MIMO channels, a SwinJSCC backbone, a simulated JPEG/BPG + LDPC baseline, and SDR hardware validation.

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Acknowledgments / References

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1. J. Huang et al., "D²-JSCC: Digital deep joint source-channel coding for semantic communications," IEEE JSAC, 2025.
2. L. X. Nguyen et al., "A contemporary survey on semantic communications: Theory of mind, generative AI, and deep JSCC," IEEE Commun. Surveys Tuts., 2025.
3. K. Yang et al., "SwinJSCC: Taming Swin Transformer for deep joint source-channel coding," IEEE TCCN, 2025.
4. Y. Diao et al., "Aligning task- and reconstruction-oriented communications for edge intelligence," IEEE JSAC, 2025.
5. M. F. Zia et al., "A survey on explainable AI for semantic communication," Comput. Netw., 2026.